

CS703 Cryptography & Information Security Study Guide

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Unit 1: Symmetric Cryptography (DES & AES)

Classical Ciphers: Caesar, Playfair, Vigenere, Hill Cipher.

Data Encryption Standard (DES):

- 64-bit block size, 56-bit effective key length (8 parity bits), 16 rounds of Feistel structure.
- Triple DES (3DES) uses 2 or 3 keys to increase key space: $C = E_{k1}(D_{k2}(E_{k3}(P)))$.

Advanced Encryption Standard (AES) - Rijndael:

- 128-bit block size with 128, 192, or 256-bit keys (10, 12, or 14 rounds).
- Substitution-Permutation Network (SPN): SubBytes, ShiftRows, MixColumns, AddRoundKey.

Unit 2: Asymmetric Cryptography & Hash Functions

RSA Algorithm (Rivest, Shamir, Adleman):

1. Choose two large prime numbers p and q .
2. Compute $n = p * q$ and Euler totient $\phi(n) = (p - 1) * (q - 1)$.
3. Choose e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
4. Compute d such that $d * e = 1 \pmod{\phi(n)}$.

Public key: (e, n) , Private key: (d, n) .

Encryption: $C = M^e \pmod{n}$. Decryption: $M = C^d \pmod{n}$.

Diffie-Hellman Key Exchange:

Allows two parties to negotiate a shared secret over an insecure channel based on discrete logarithm difficulty.

Hash Functions: SHA-256 and MD5 provide collision resistance and pre-image resistance for digital signatures and HMAC.

